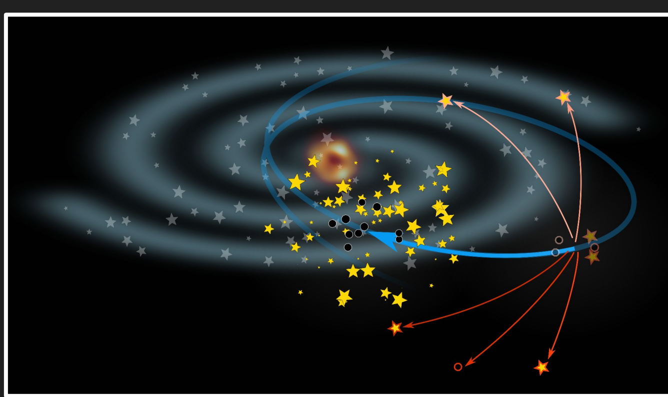
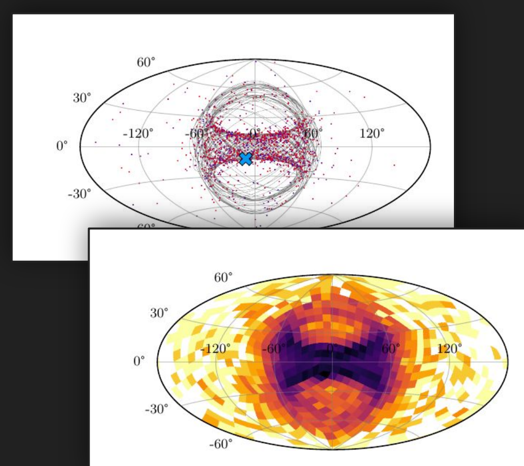
Runaway and Hypervelocity Stars From Compact Object Encounters in Globular Clusters

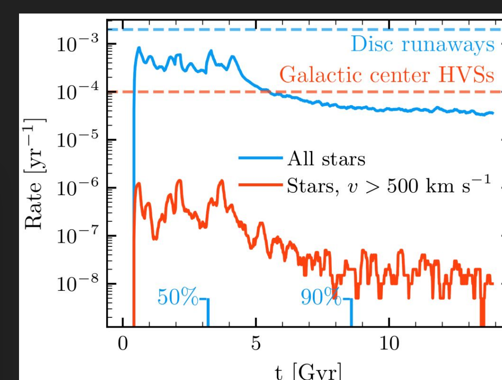
Tomás Cabrera | Carnegie Mellon University

MODEST-23 | August 28, 2023

Paper: [Cabrera+Rodriguez23]







Motivation

THE ASTRONOMICAL JOURNAL VOLUME 101, NUMBER 2 FEBRUARY 1991

THE MAXIMUM POSSIBLE VELOCITY OF DYNAMICALLY EJECTED RUNAWAY STARS

PETER J. T. LEONARD

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Received 16 May 1990; revised 8 October 1990

- ***How much faster/frequent*** are ejecta from encounters involving compact objects?

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THE ASTROPHYSICAL JOURNAL, 866:121 (11pp), 2018 October 20 https://doi.org/10.3847/1538-4357/aadee5

CrossMark

Old, Metal-poor Extreme Velocity Stars in the Solar Neighborhood*

Kohei Hattori[1], Monica Valluri[1], Eric F. Bell[1], and Ian U. Roederer[1,2]
[1] Department of Astronomy, University of Michigan, 1085 S. University Ave., Ann Arbor, MI 48109, USA; khattori@umich.edu
[2] Joint Institute for Nuclear Astrophysics—Center for the Evolution of the Elements (JINA-CEE), USA
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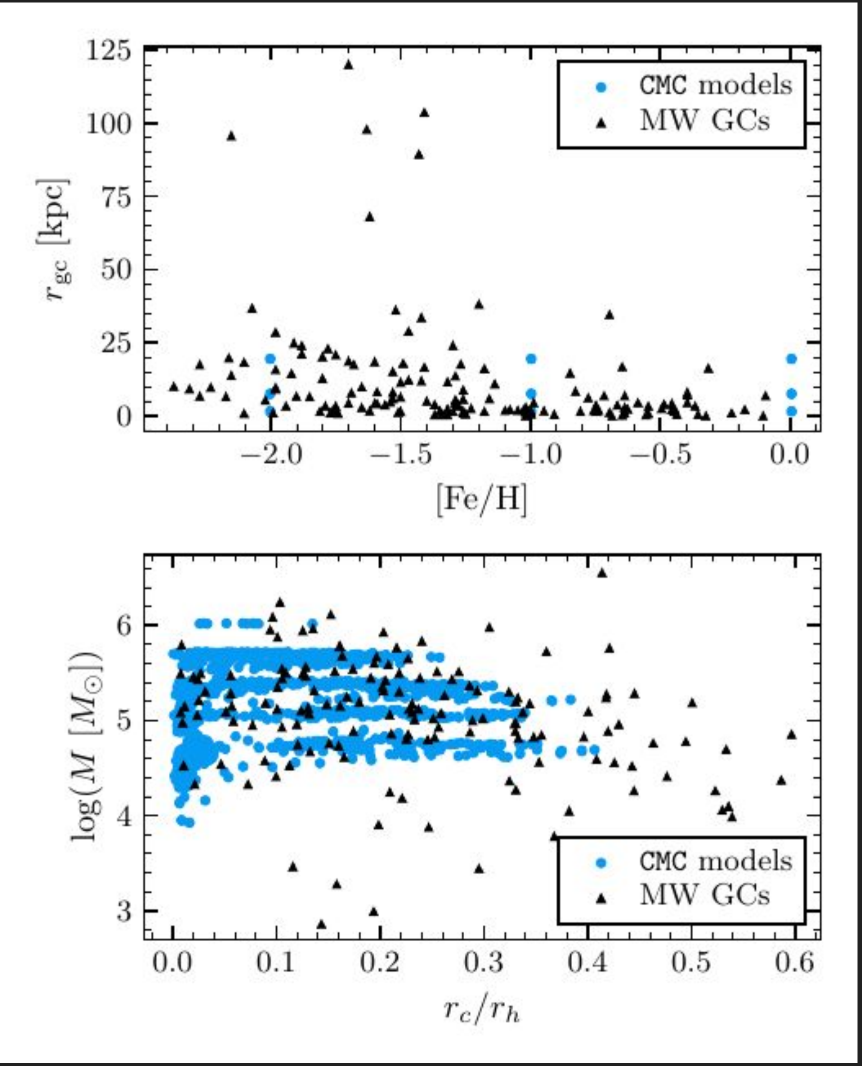
- To what degree ***do globular cluster ejections contribute*** to the population of runaway/hypervelocity stars?

Matching CMC models to Milky Way GCs

Observational catalogs:

- Baumgardt+Hilker18 (masses, radii, galactic trajectories)
- Harris10 (metallicities)
- **149 GCs covered by both**



Matching CMC models to Milky Way GCs

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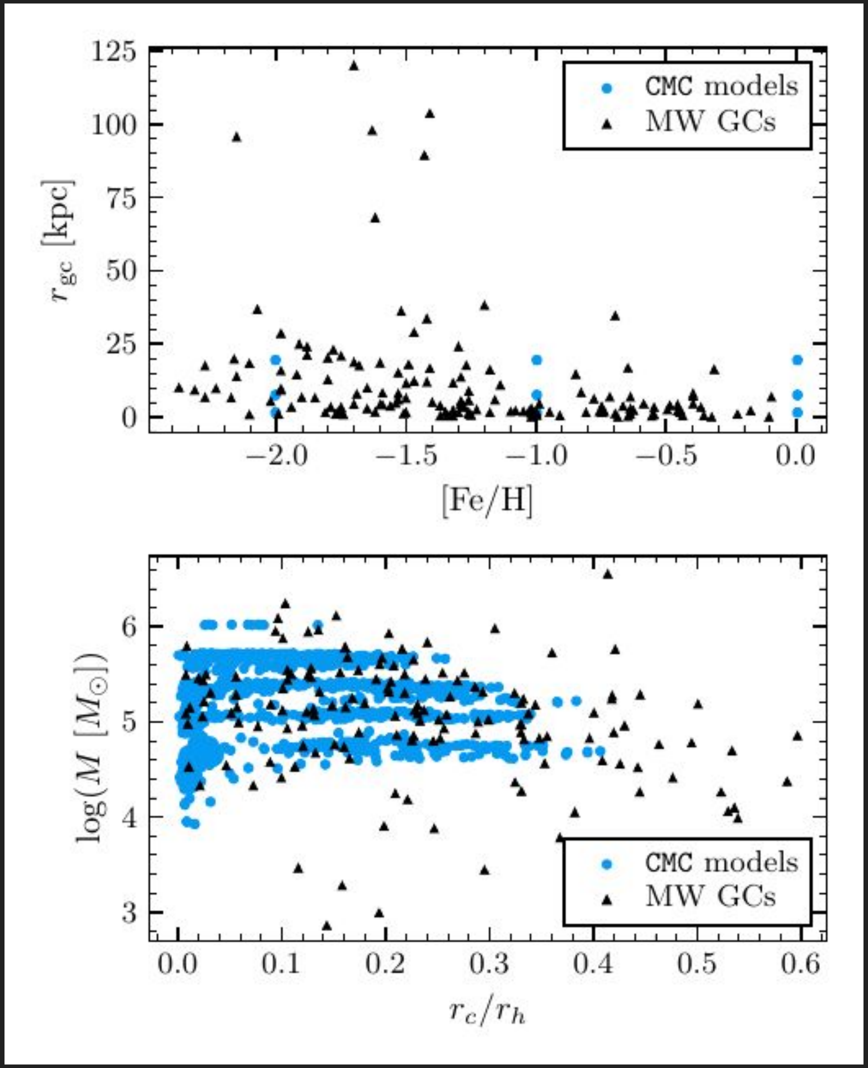
Simulated catalog:

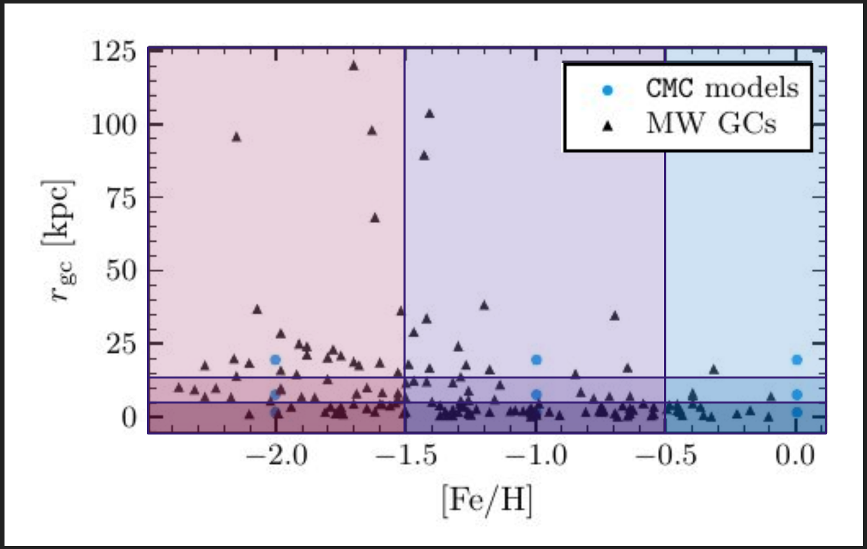
- Cluster Monte Carlo (CMC) Cluster Catalog (Kremer+20b)
- **148 GC models spanning MW GC parameter space**



Matching CMC models to Milky Way GCs

Matching:

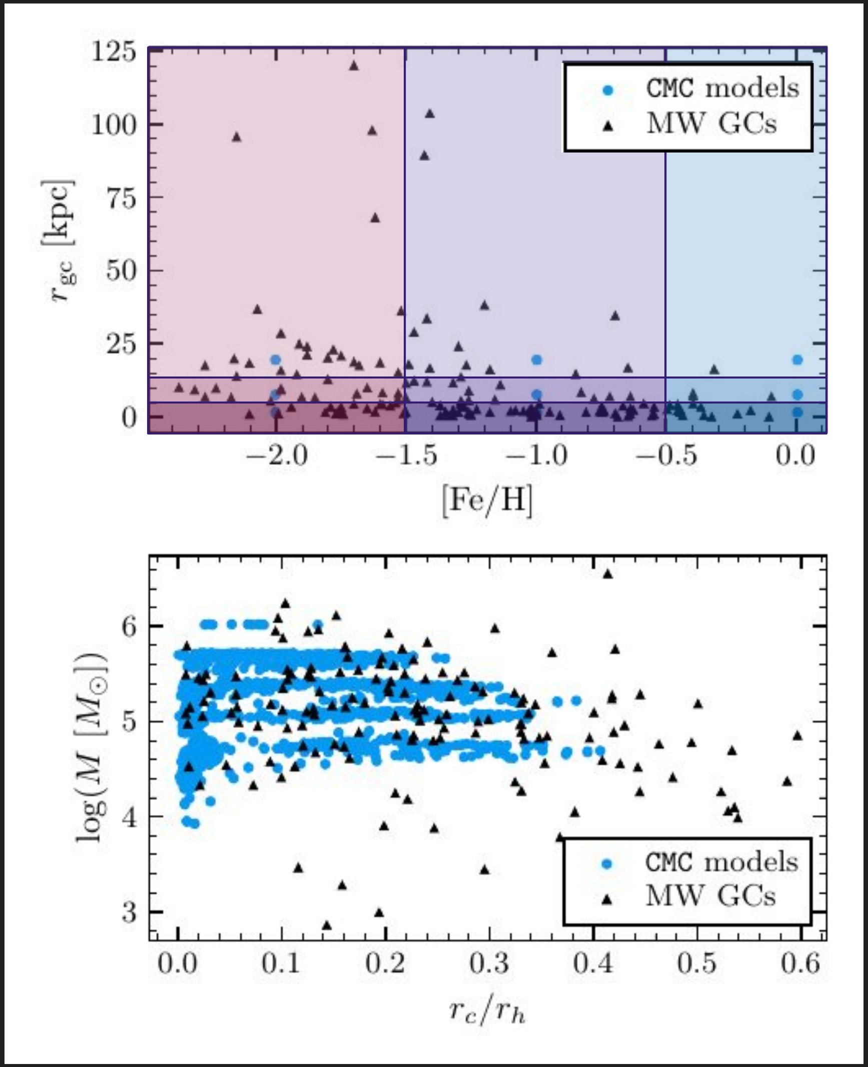
- Choose **closest point in (r_{gc}, [Fe/H])** parameter grid



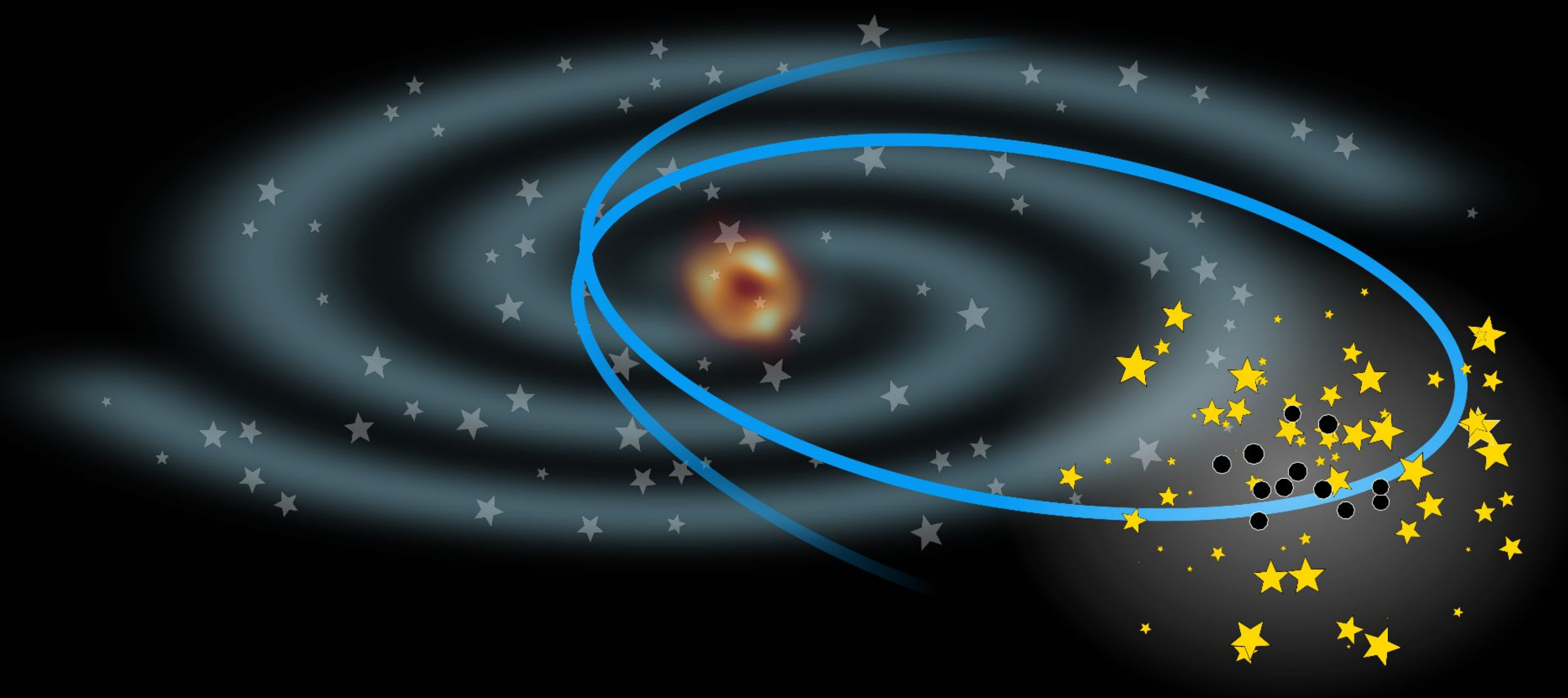
Matching CMC models to Milky Way GCs

Matching:

- Choose **closest point in (r_{gc}, [Fe/H])** parameter grid
- Out of those models, choose **closest snapshot in (log M, r_c/r_h)** space, out of snapshots between 10-13.5 Gyr



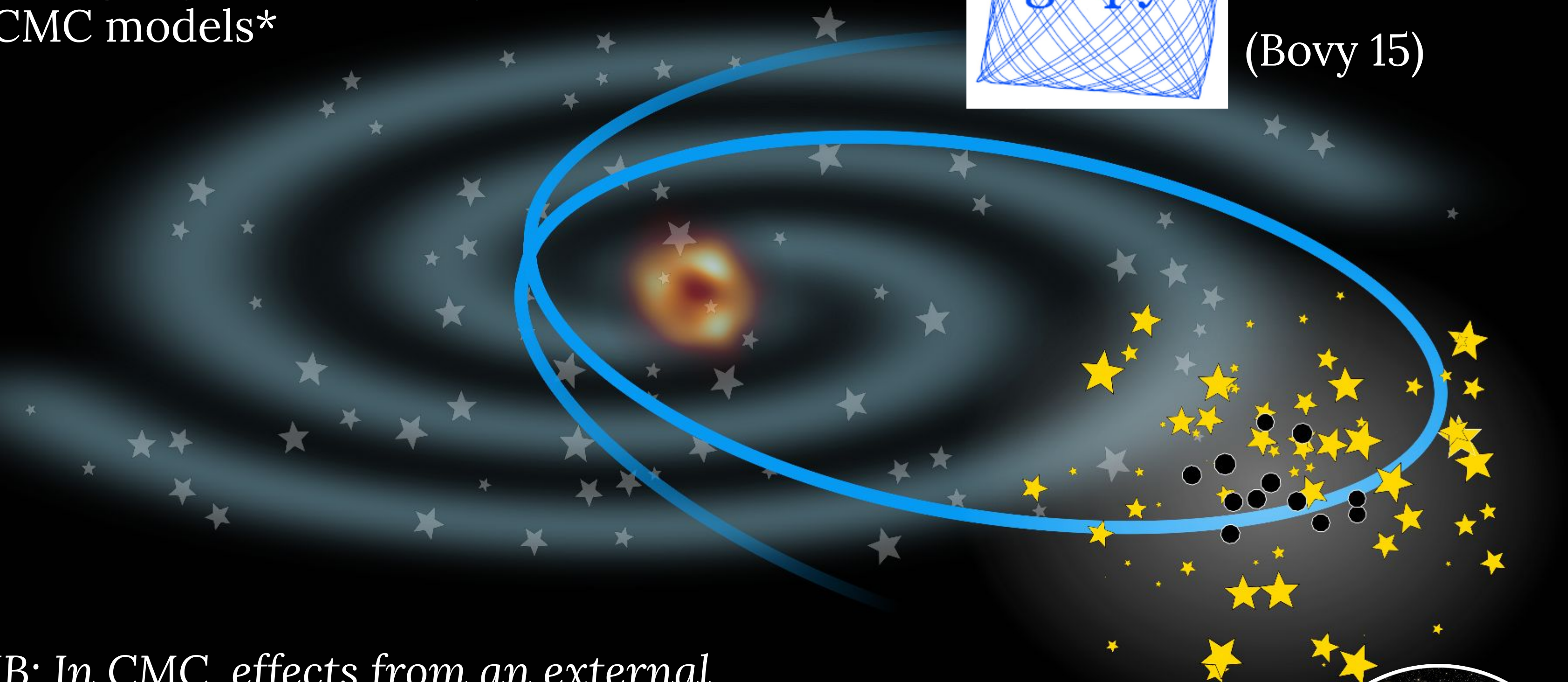
Simulating runaway stars



Sag. A* image: https://www.eso.org/public/images/eso2208-eht-mwa/

Simulating runaway stars

1. **Integrate GC orbit** in MW potential with galpy, **follow GC dynamics** in CMC models*



(Bovy 15)

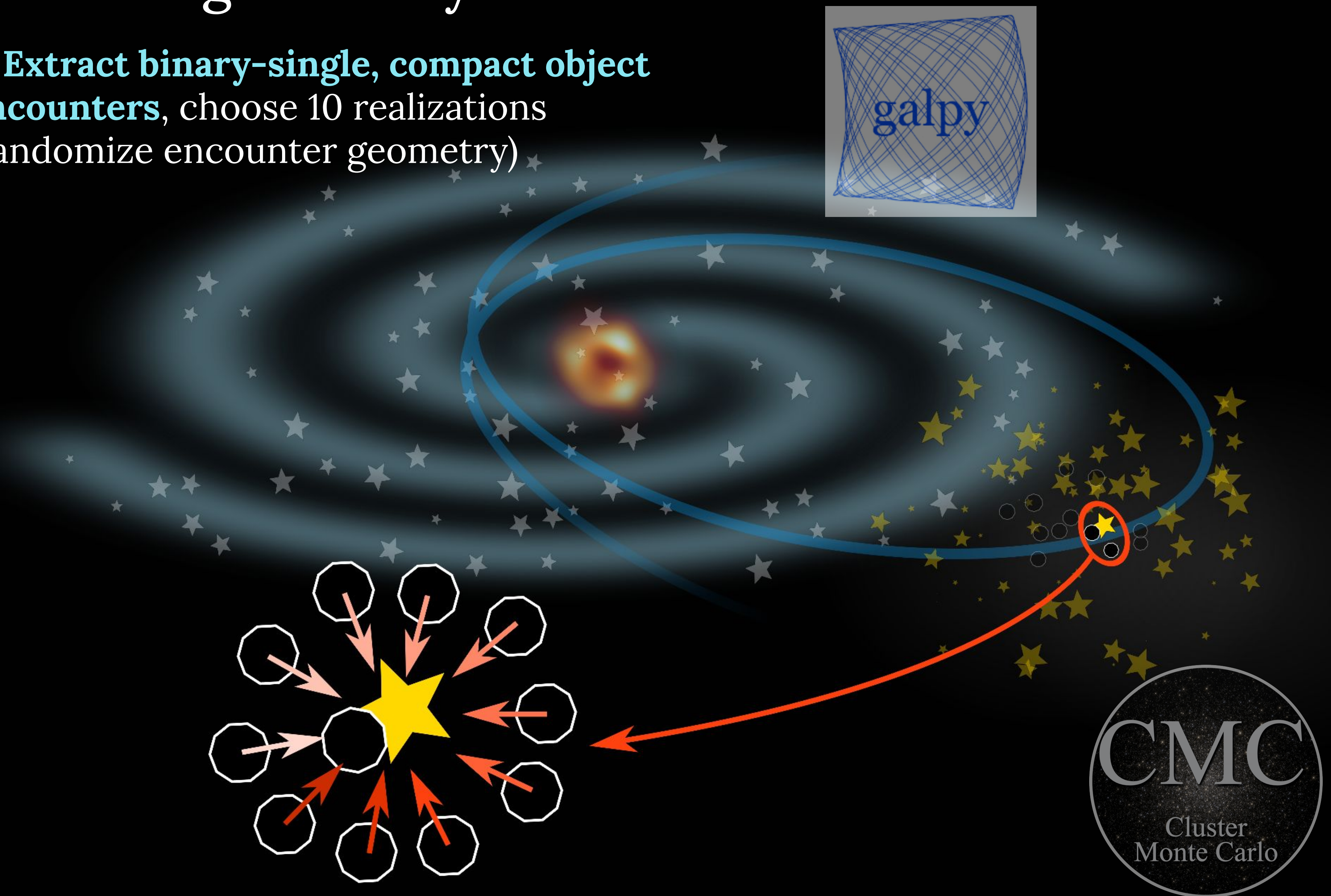
** – NB: In CMC, effects from an external potential are static and spherically symmetric i.e. internal GC dynamics are computationally decoupled from galactic orbital evolution*

(Joshi+00, Pattabiraman+13, Rodriguez+22)

CMC Cluster Monte Carlo

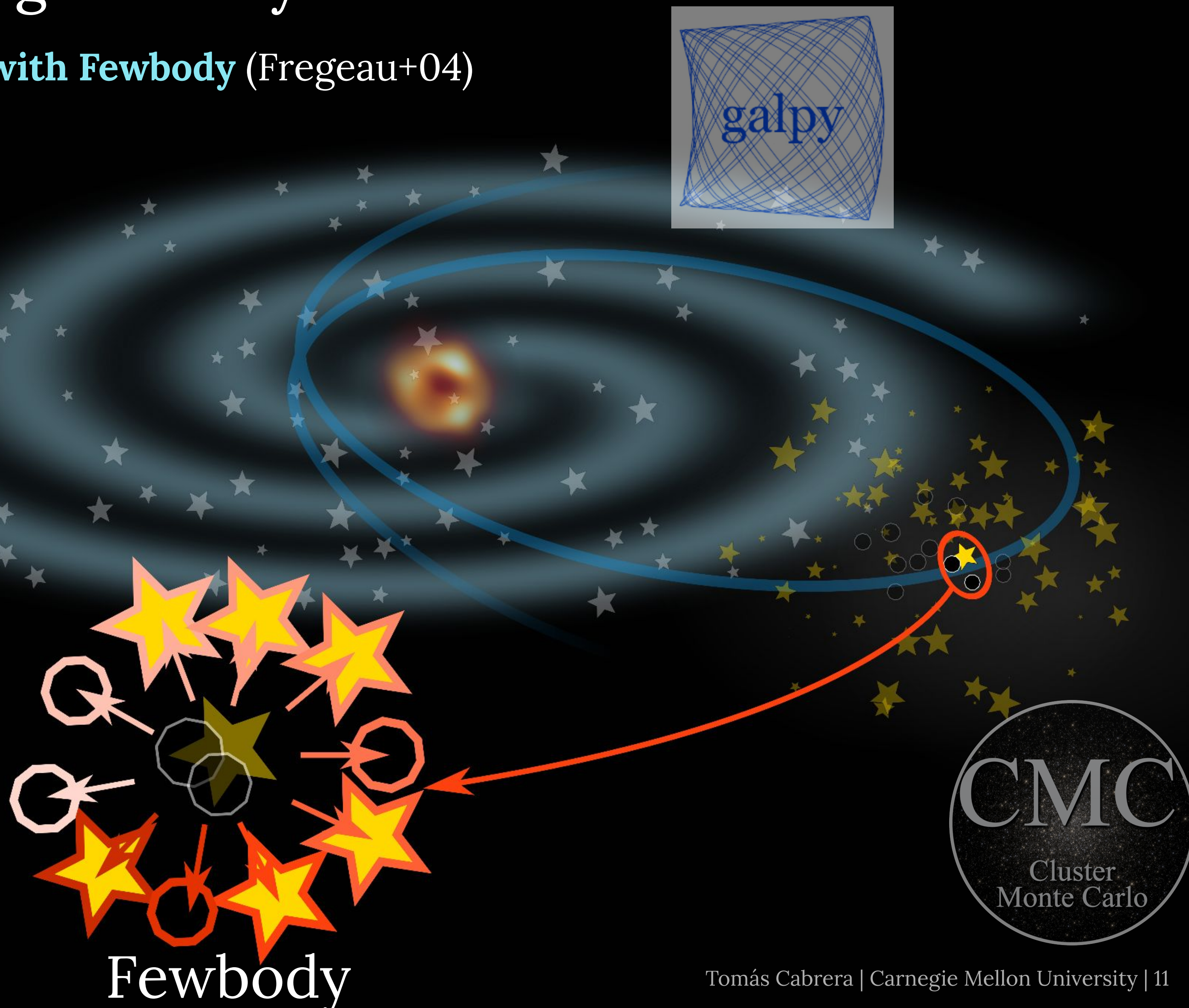
Simulating runaway stars

2. **Extract binary-single, compact object encounters**, choose 10 realizations (randomize encounter geometry)

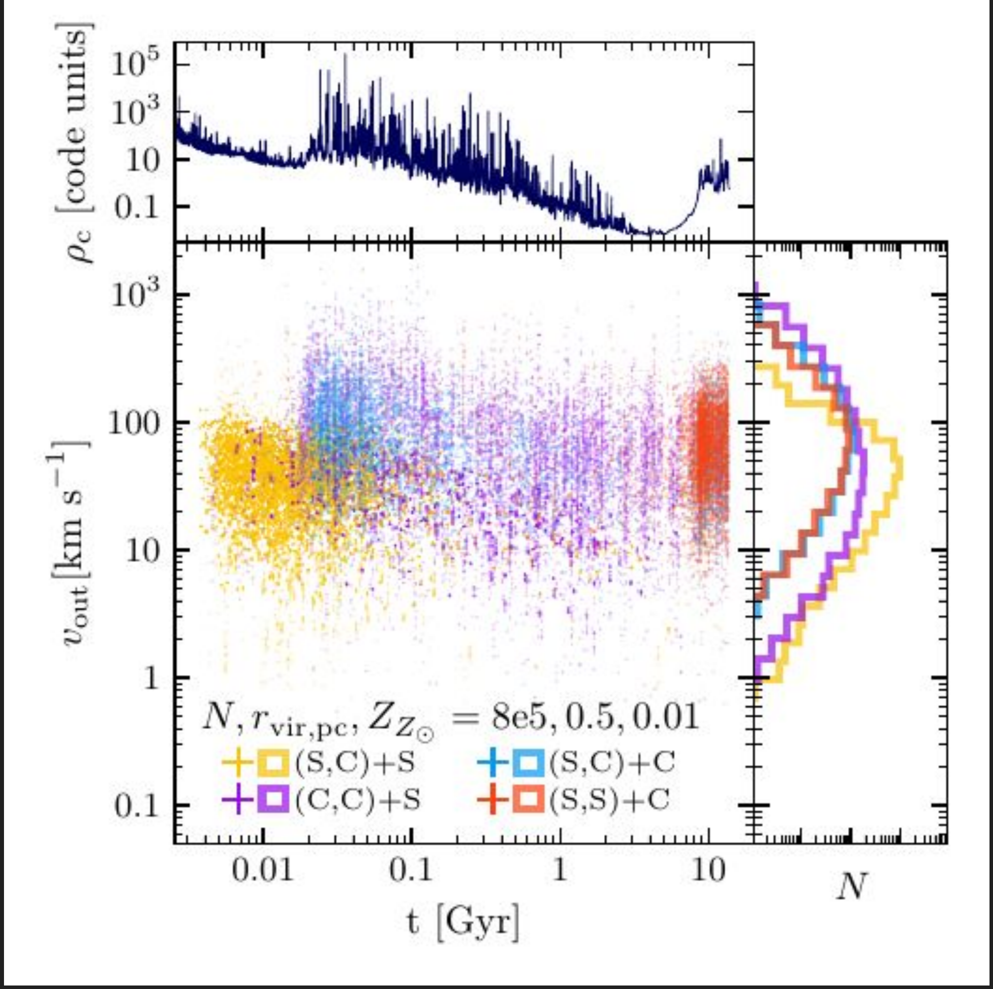


Simulating runaway stars

3. **Integrate with Fewbody** (Fregeau+04)



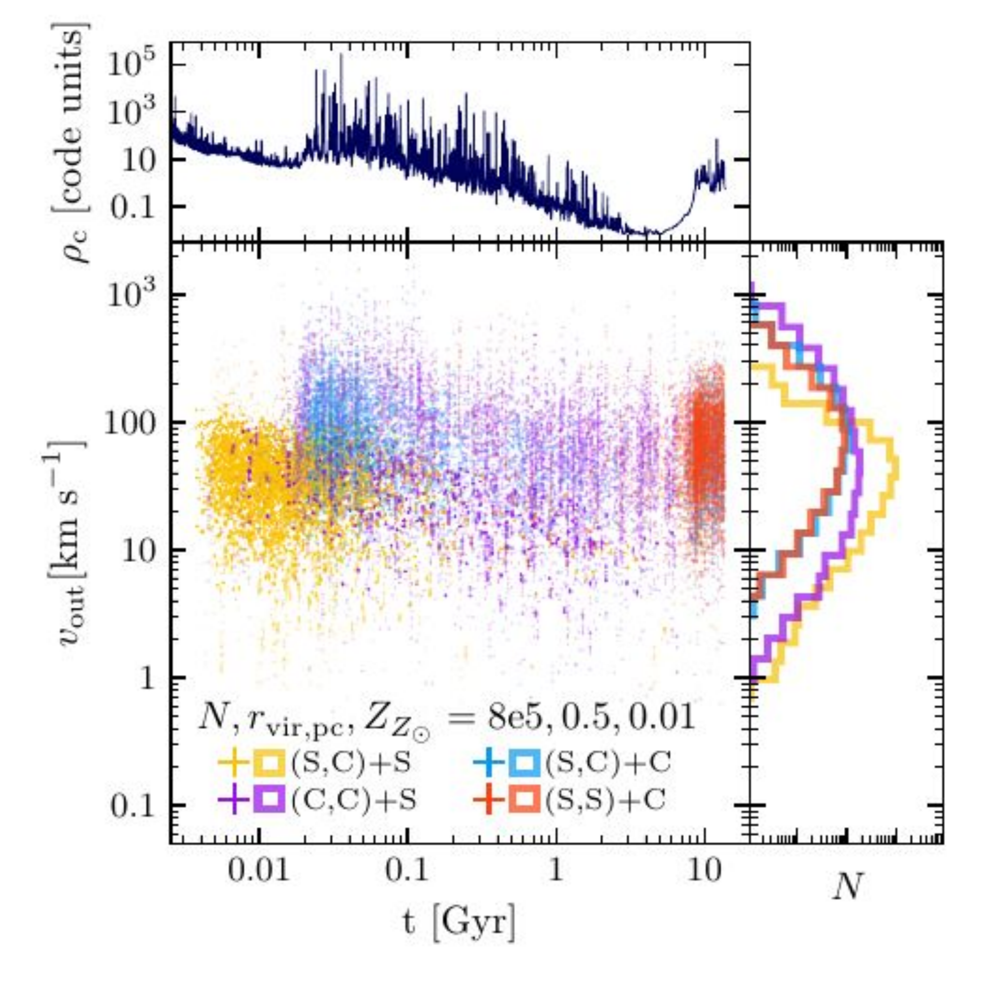
Ejecta populations as a function of model evolution



Fiducial model:

- $N = 8e5$
- $r_{vir} = 0.5$ pc
- $r_{gc} = 8$ kpc
- $Z/Z_\odot = 0.01$

Ejecta populations as a function of model evolution



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Notes:

- Synchronization with **core collapses**

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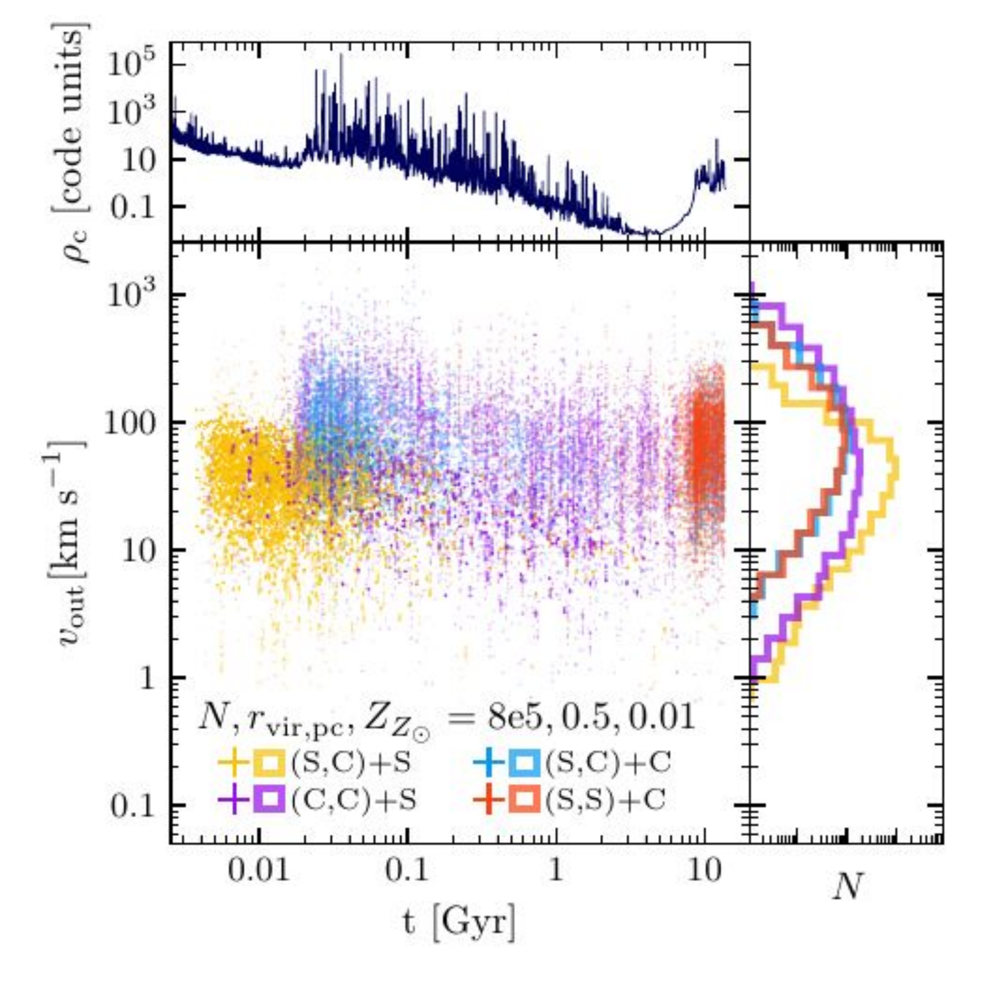

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- Different encounter compositions (colors)

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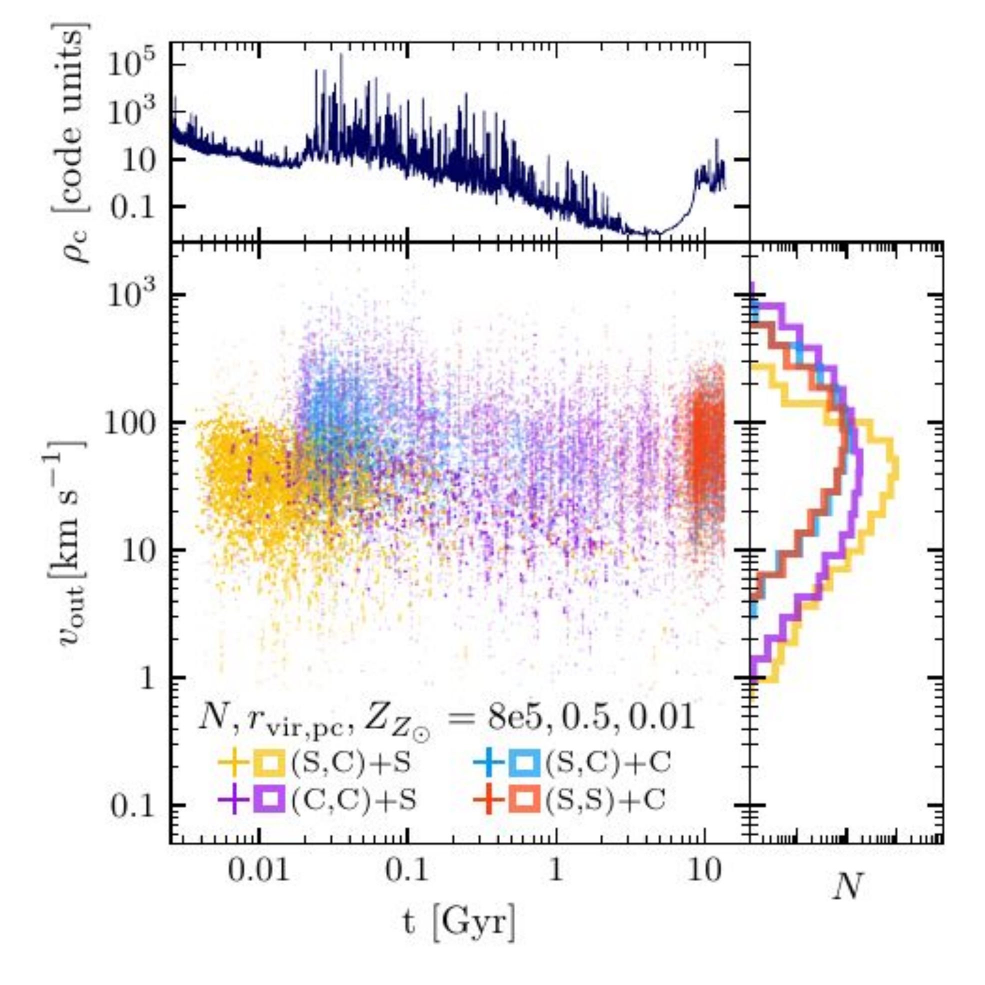

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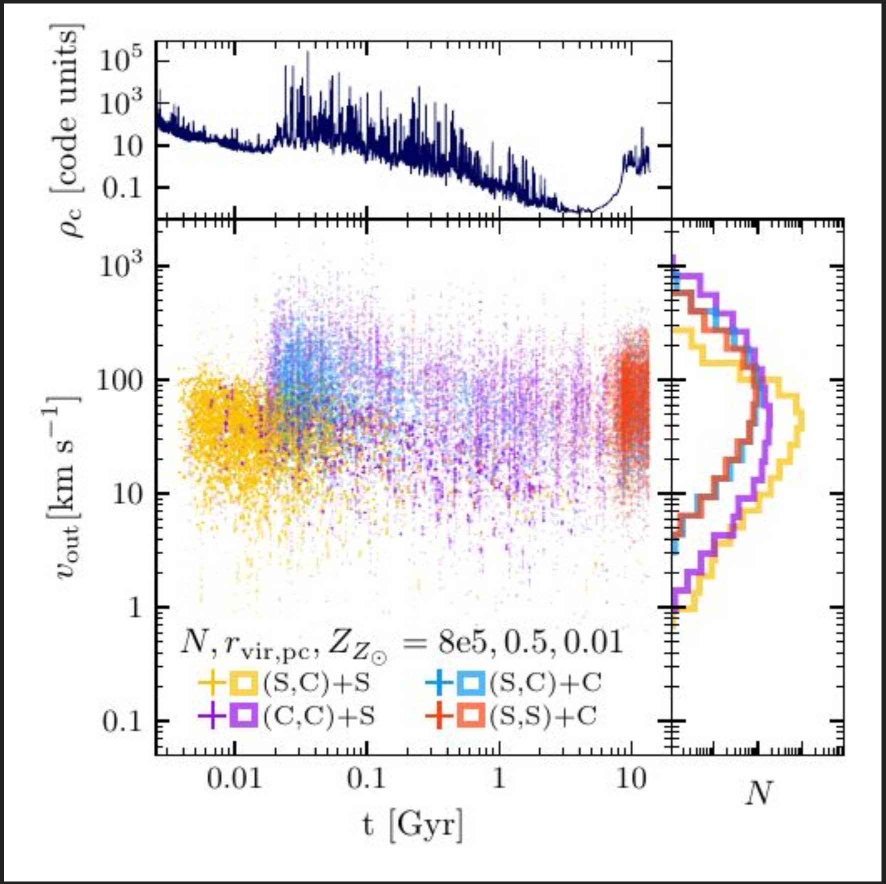
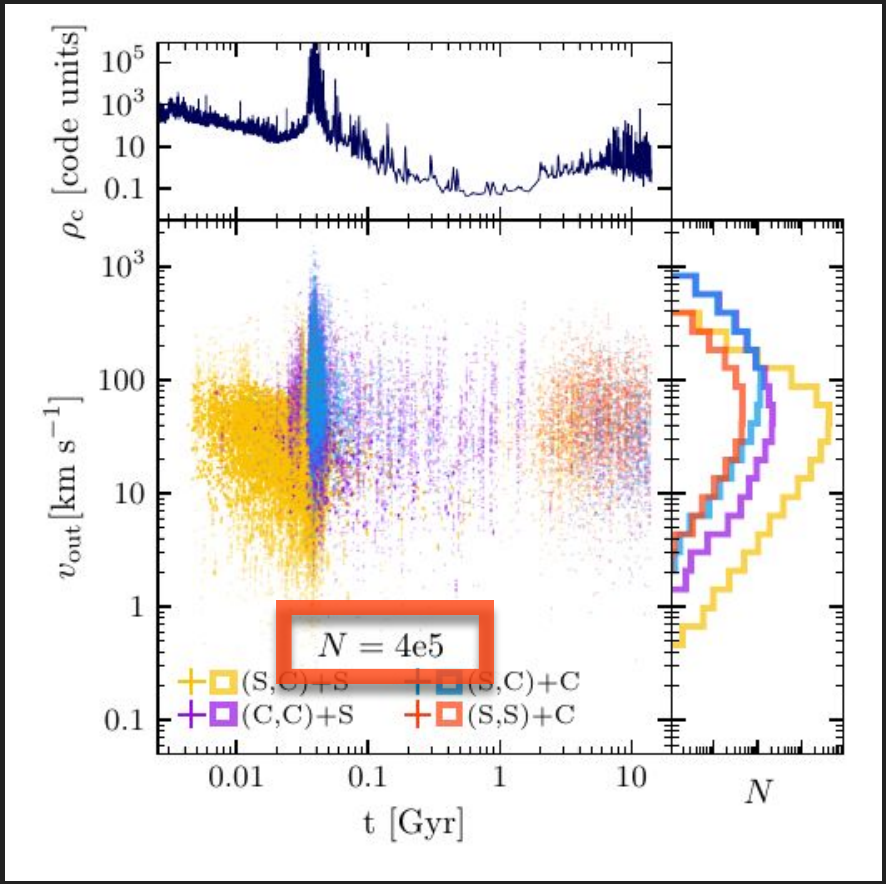
Notes:

- Synchronization with **core collapses**
- Different encounter compositions (colors)
- After second "BH-burning" core collapse (e.g. Kremer+20a), **WDs (+NSs)** dominate encounters

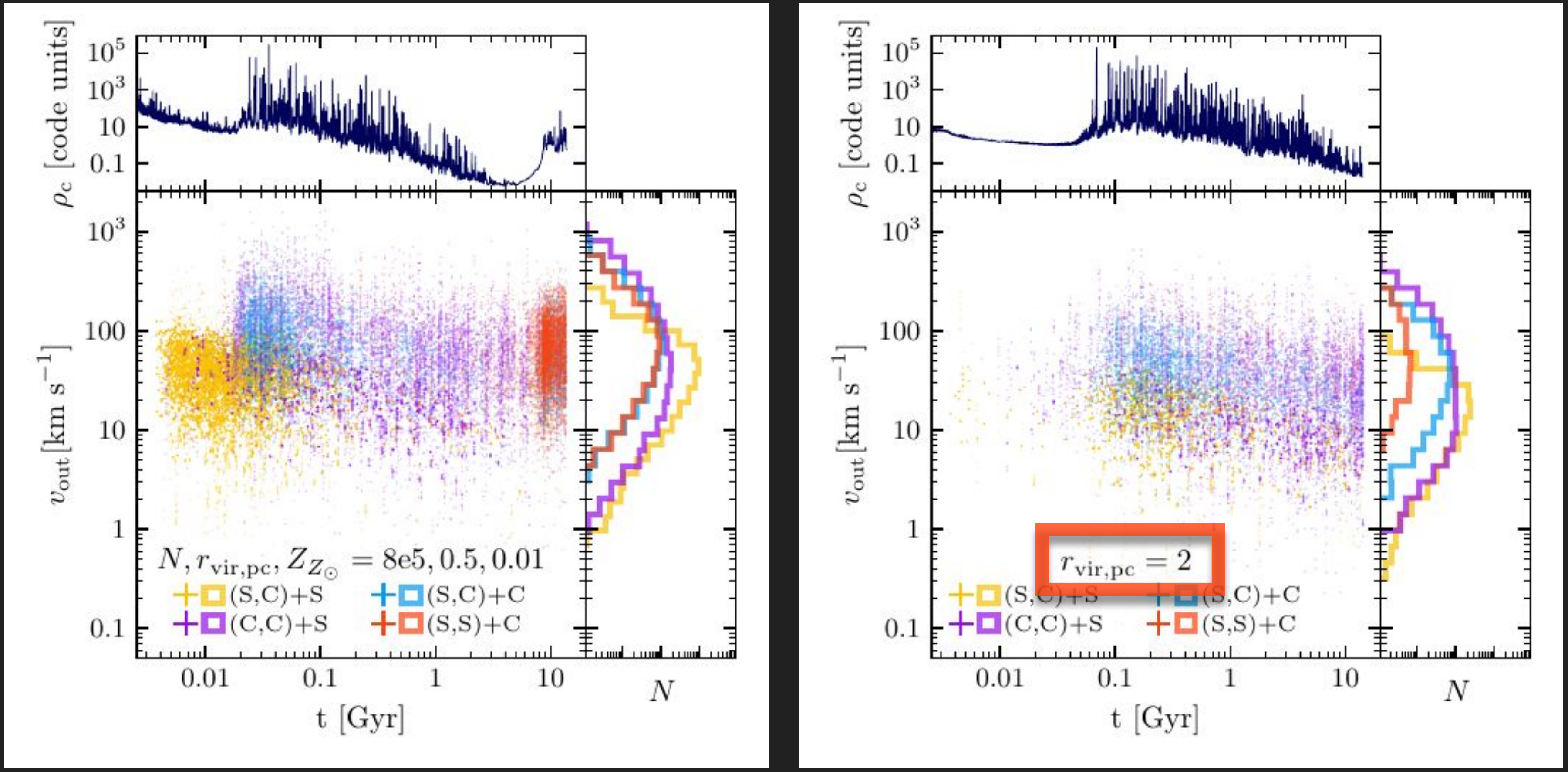
Ejecta populations as a function of model parameters





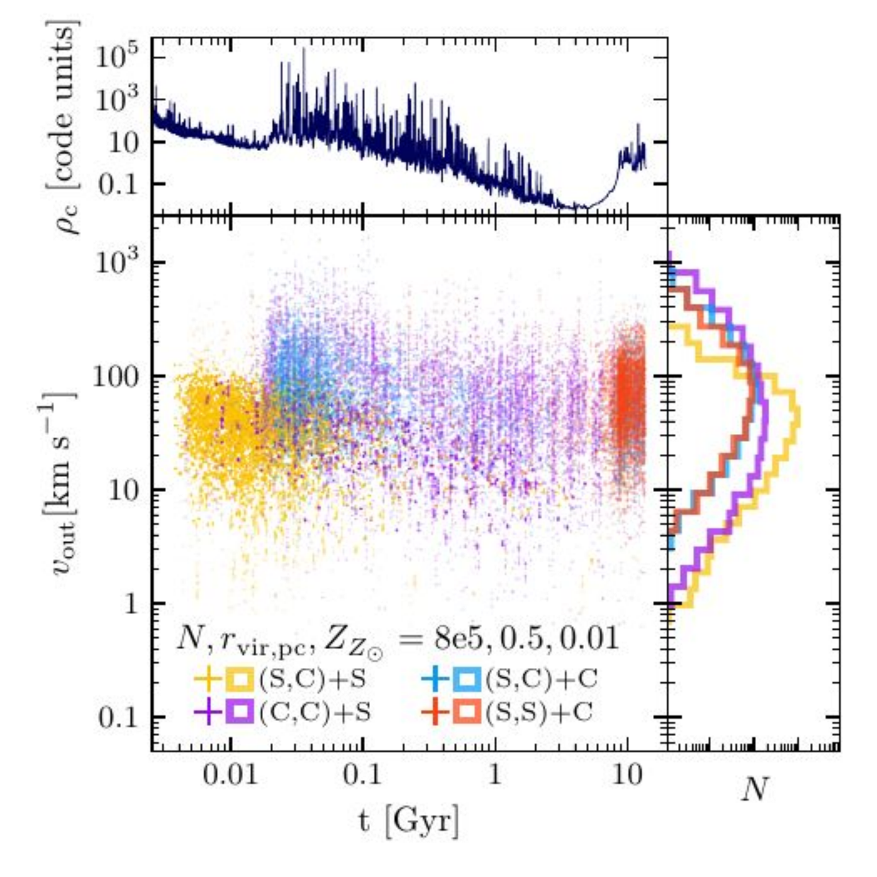
- Clusters with ***fewer*** objects can have ***more, stronger*** ejections (weaker subsystem -> more severe core collapse)

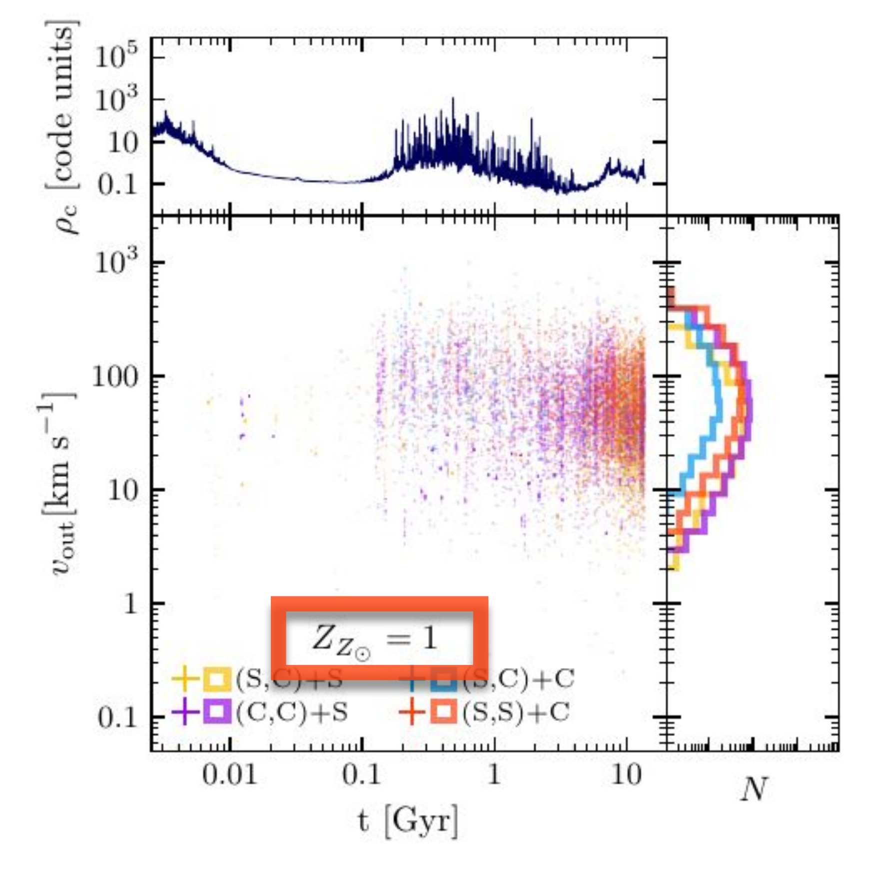
Ejecta populations as a function of model parameters



- ***Larger*** clusters have ***fewer, weaker*** ejections
  (slower relaxation time -> later, slower, less severe core collapse)

Ejecta populations as a function of model parameters

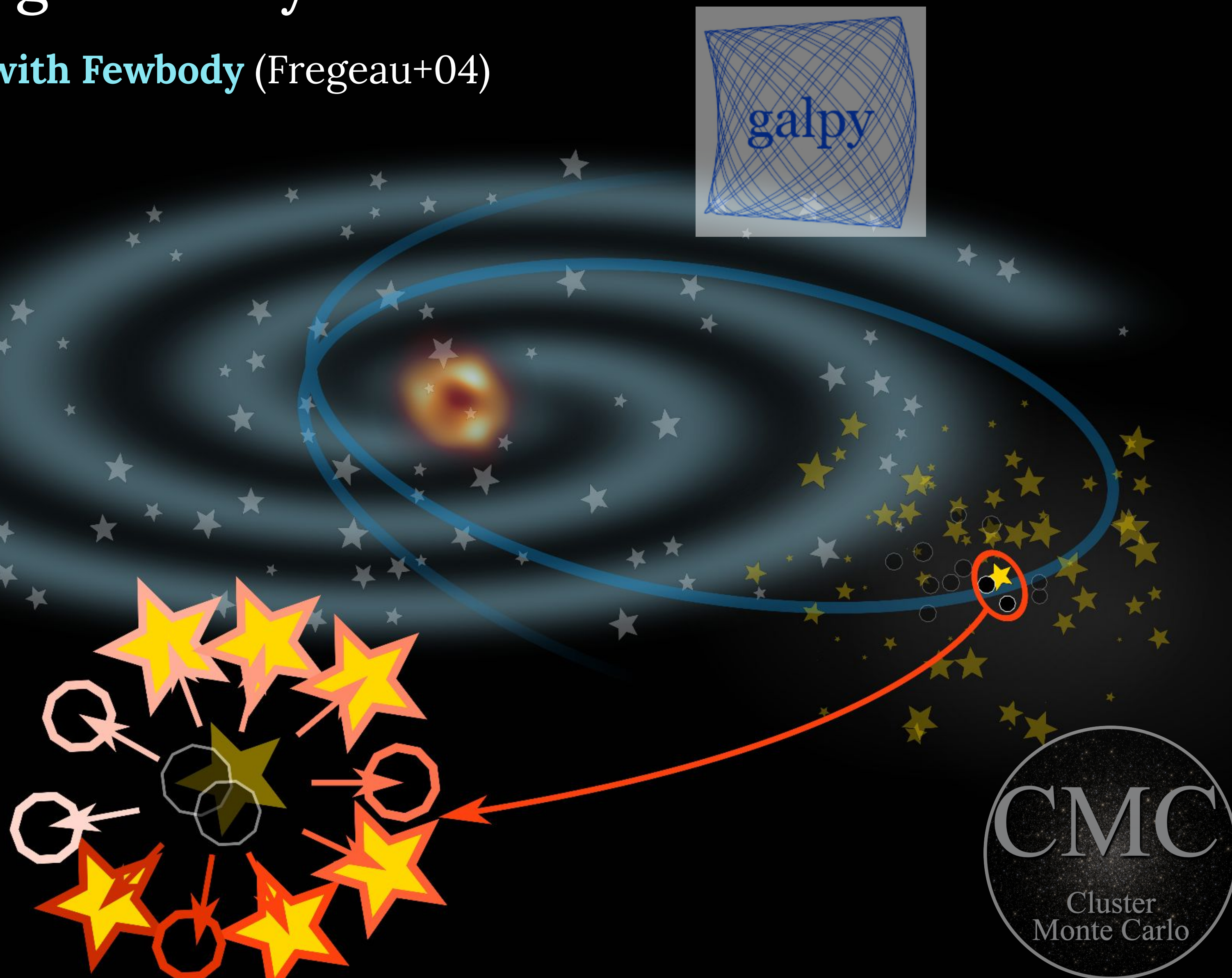




- ***High-metallicity*** clusters have ***fewer, weaker*** ejections (lower-mass, later, fewer BHs -> less severe core collapse)

Simulating runaway stars

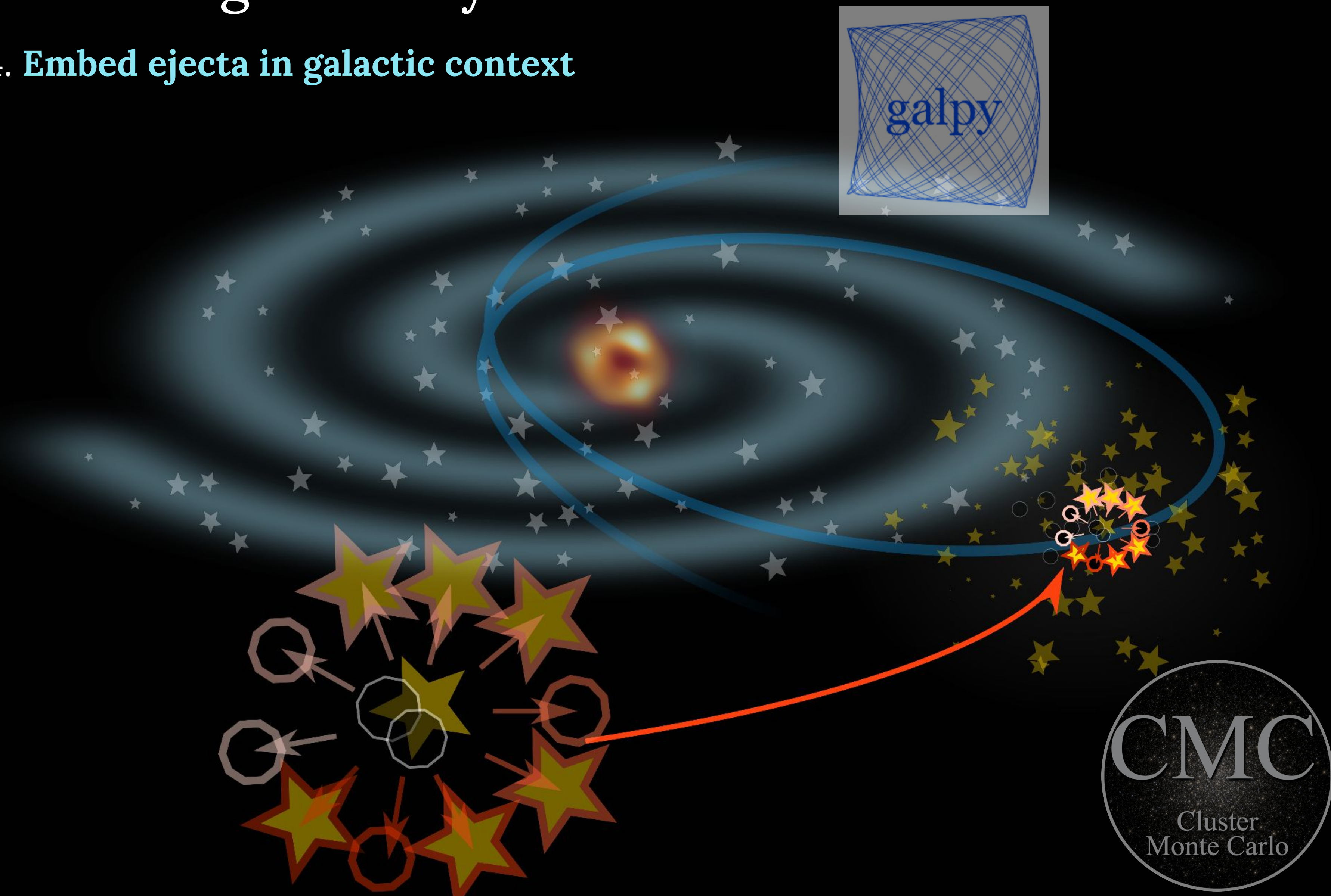
3. **Integrate with Fewbody** (Fregeau+04)



Fewbody

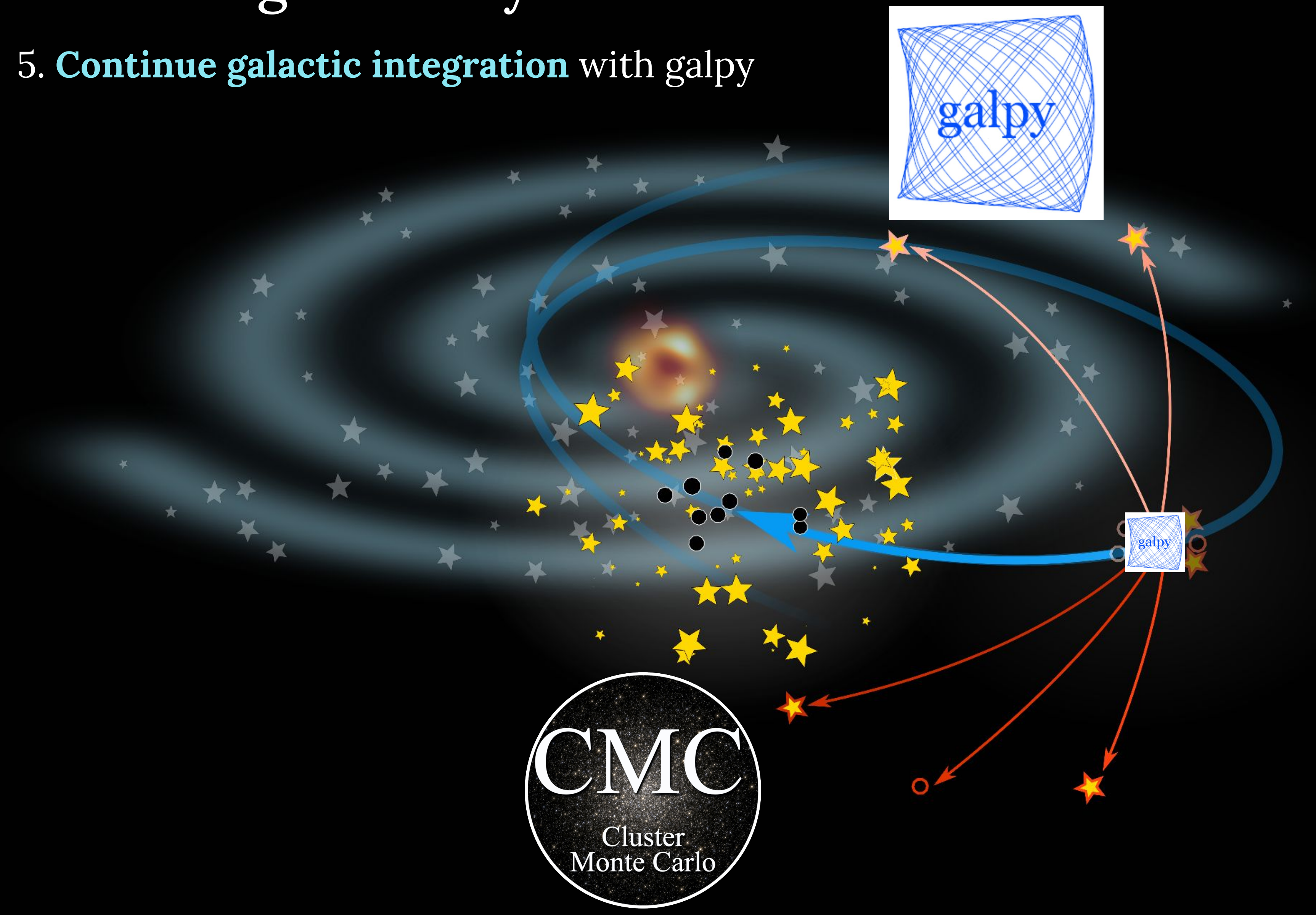
Simulating runaway stars

4. **Embed ejecta in galactic context**



Simulating runaway stars

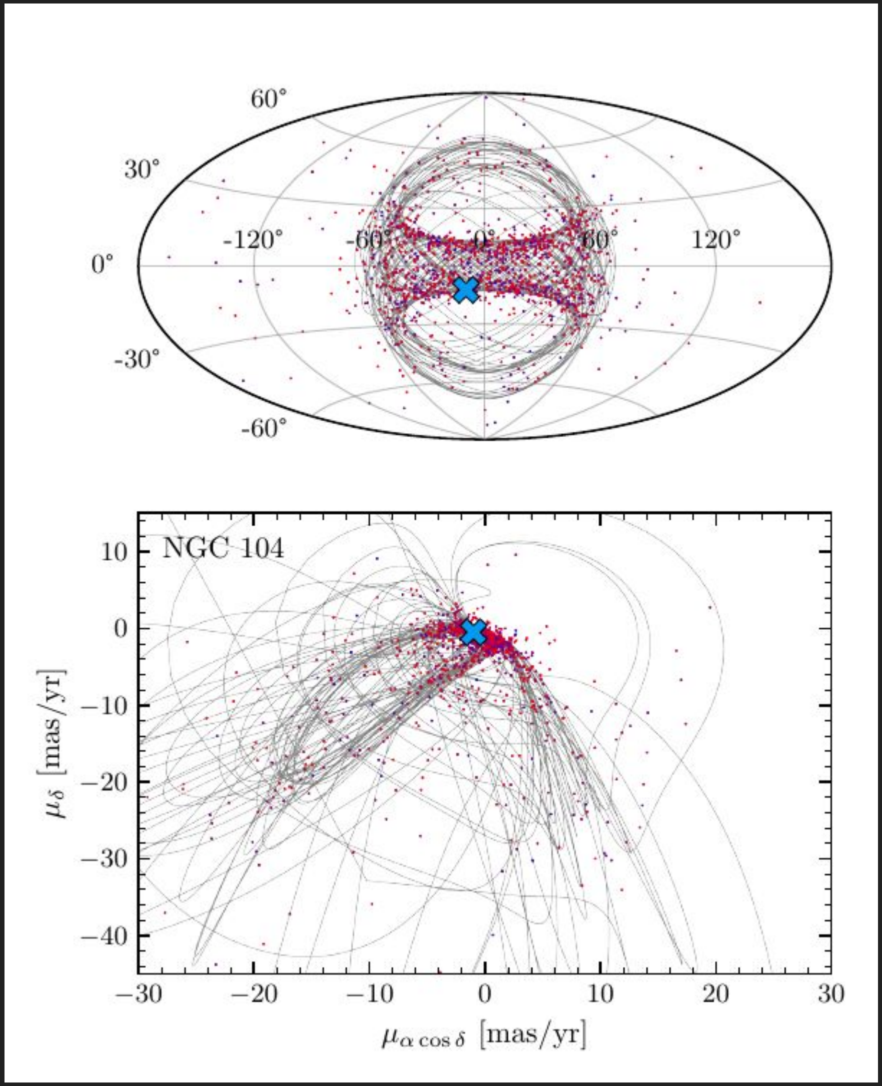
5. **Continue galactic integration** with galpy



Phase space distributions of ejecta

https://doi.org/10.5281/zenodo.7599870

1 distribution :: 1 Milky Way GC (149 total)



Available as:

- **Full list of ejecta** (with 3D positions/velocities, masses, stellar types, etc.)

Phase space distributions of ejecta

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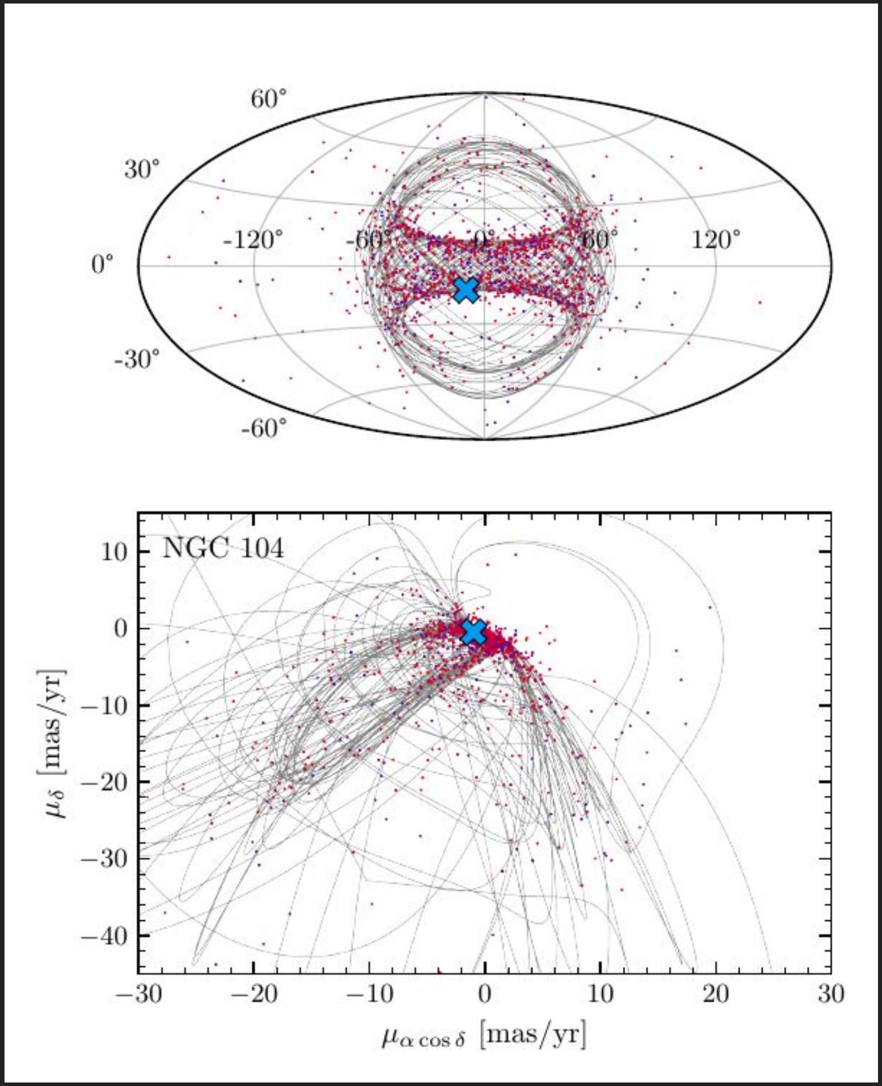
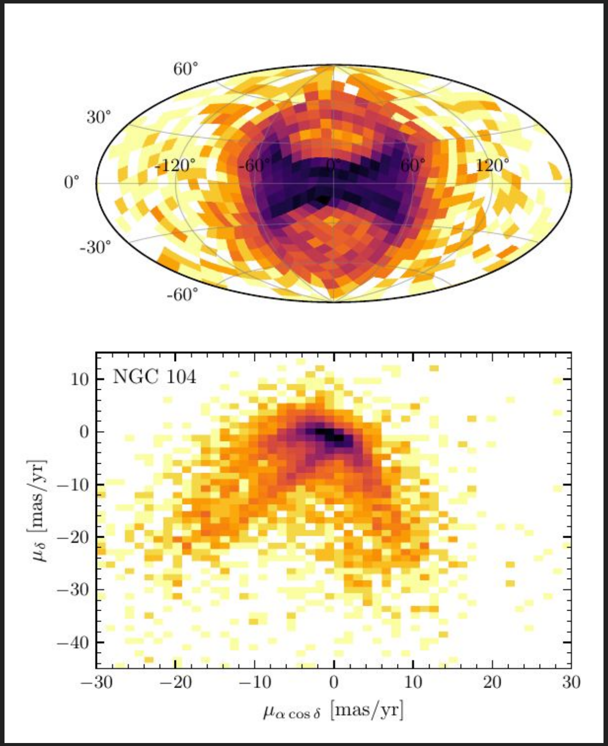
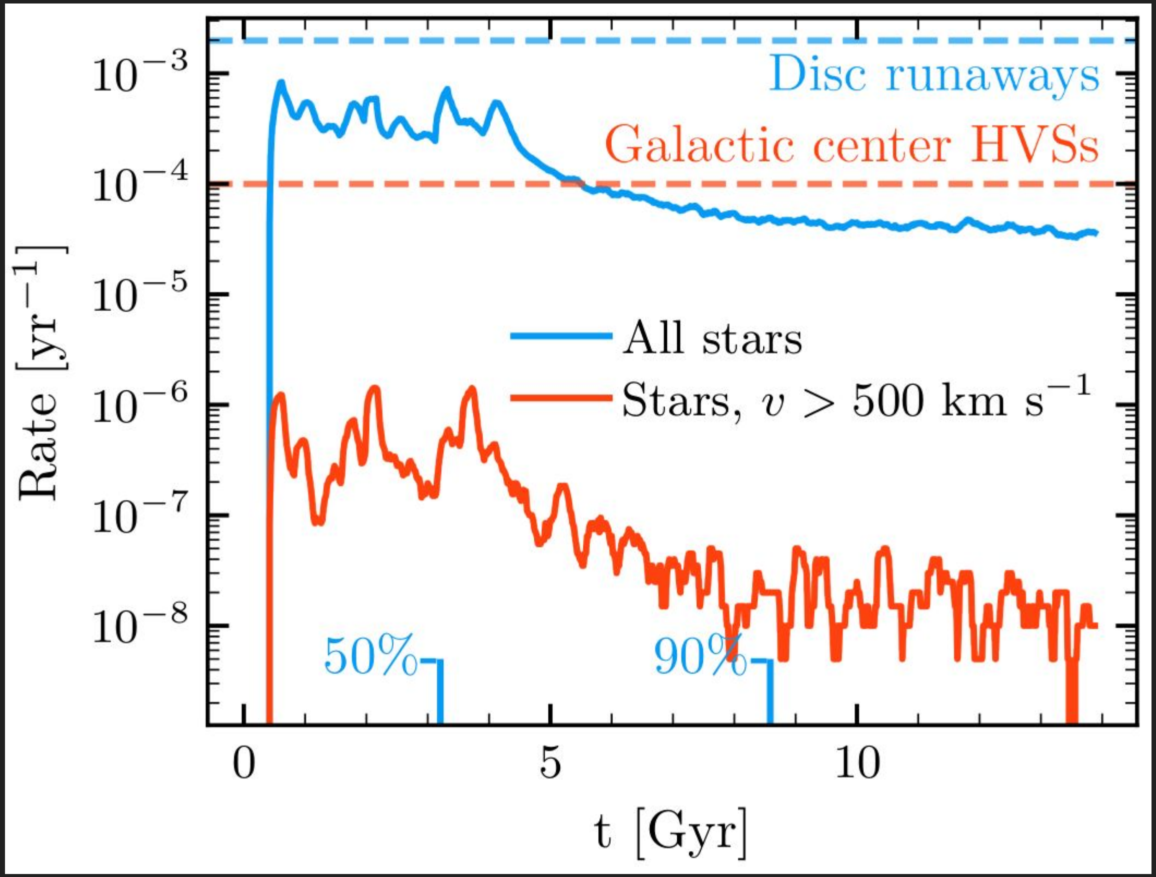




Available as:

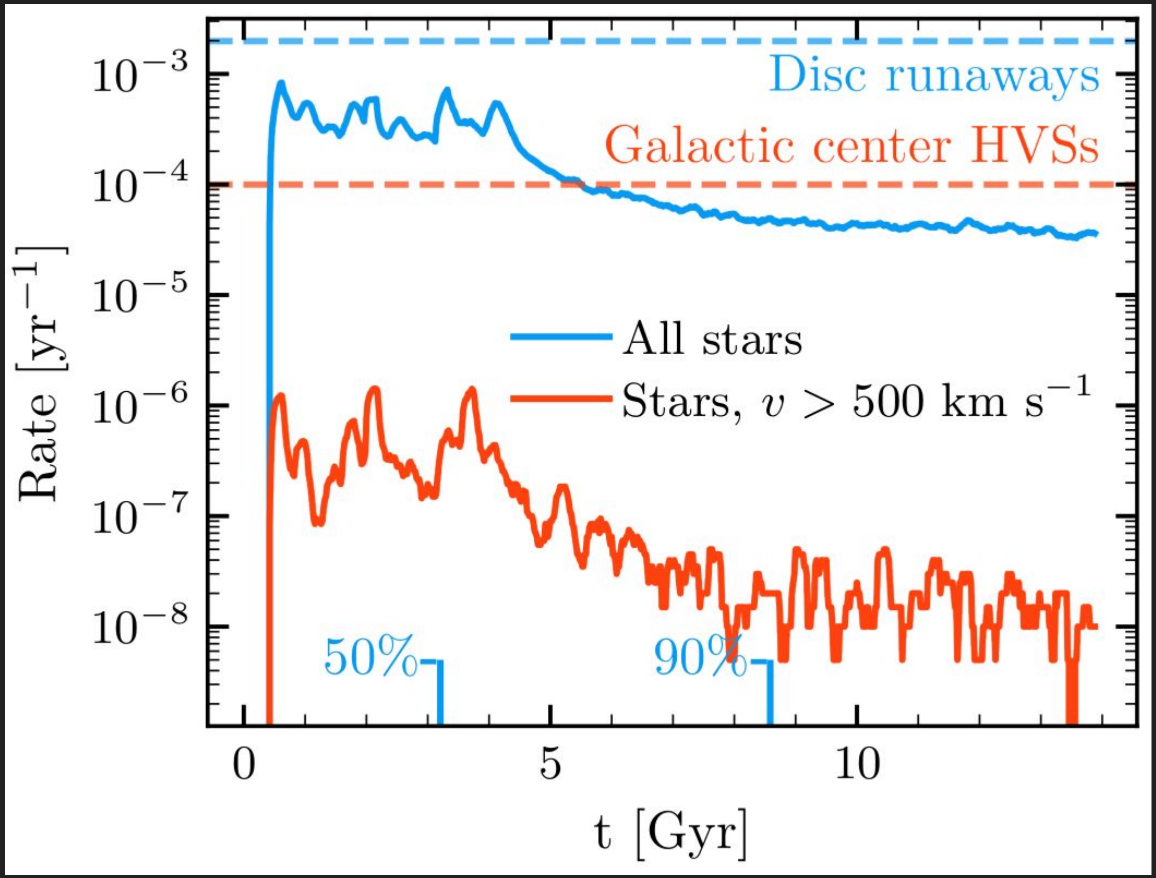
- **Full list of ejecta** (with 3D positions/velocities, masses, stellar types, etc.)
- **2D histograms** (sky position is "HEALPix-ellated")

Galactic encounter rates



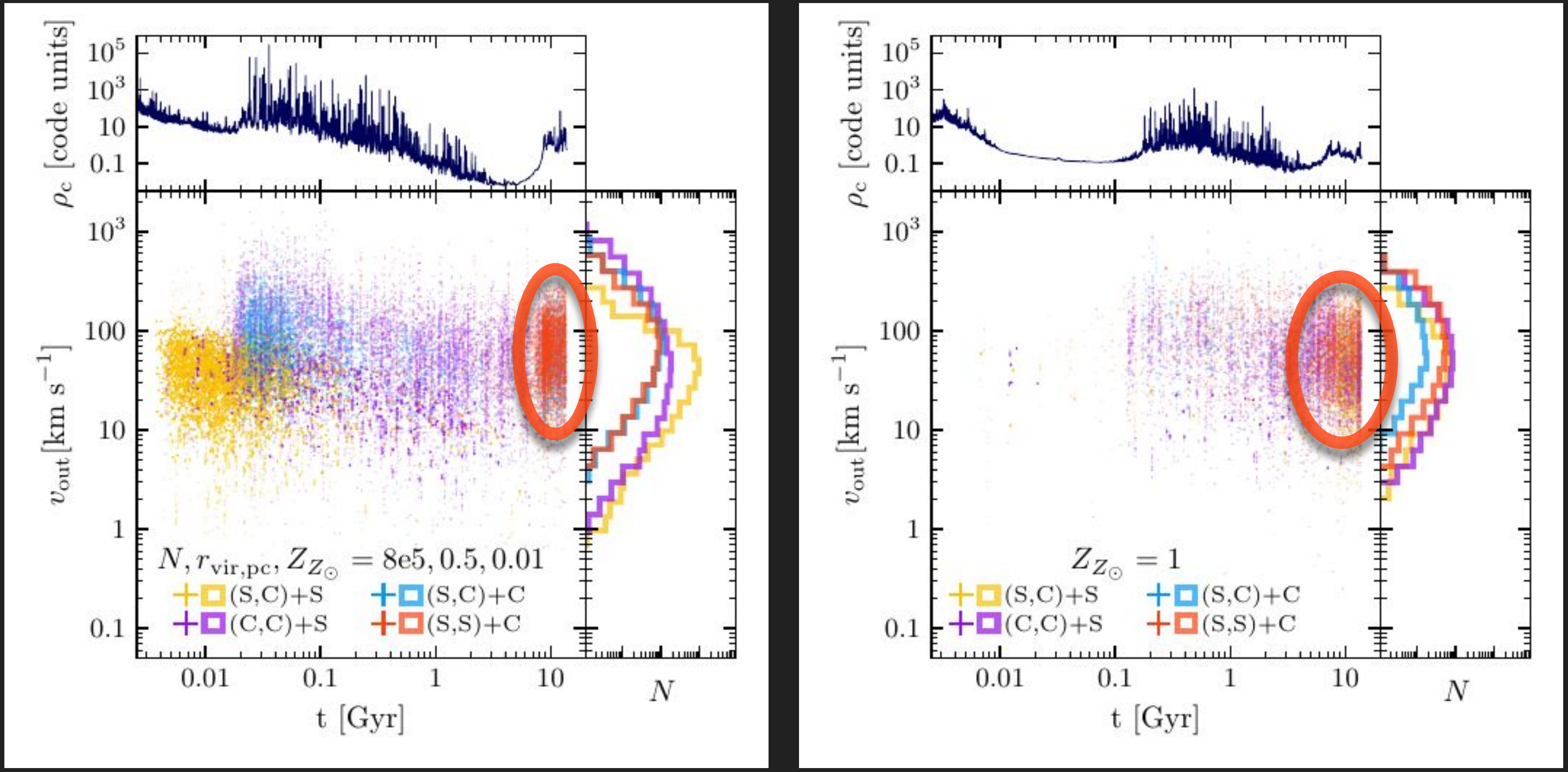
- **Ejections peak in the first few Gyr** after GC formation

Galactic encounter rates



- **Ejections peak in the first few Gyr** after GC formation
- **Hypervelocity stars produced**, but at much lower rates than the galactic center (Hills mechanism, e.g. Brown15)

Runaway WDs?



- After BH depletion core collapse, WDs concentrate in core -> **more WD ejections?**

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